A. Per-head update step

k_h v_h q_h d_h g_h e_h w_h

$\downarrow$ L^2 norm q, k $\downarrow$

$$r_h = S_h^T(e_h \odot k_h) \quad \text{(gated read at address } e_h \odot k_h\text{)}$$

$\downarrow$

$$\delta_h = w_h \odot v_h - r_h \quad \text{(prediction error)}$$

$\downarrow$

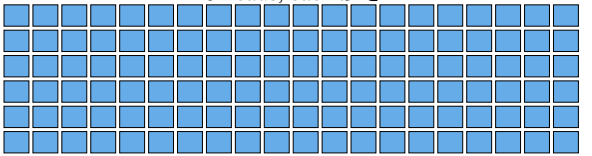
$$S'_h = \tanh\left(d_h S_h + k_h \delta_h^T\right)$$

$\downarrow$

$$y_h = \text{silu}(g_h) \odot (S'_h)^T q_h$$

B. Multi-programmed shape (per layer, per batch element)

$H = 216$ heads, each $S \in \mathbb{R}^{32\times 32}$



··· 216 small recurrent programs per layer ···

At batch size 4 and depth 11:

9,504 independent programs per token

Each program: a 32×32 state tile in registers/SRAM; serial along time; parallel across $H \times$ batch.